
Test-Time Refinement Moves Intelligibility Six Times More Than Identity in Masked-Diffusion TTS

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 A diffusion language model buys sequential computation twice: depth, paid once
2 in parameters, and denoising steps, rented per utterance at serve time. We ask
3 whether the rented kind helps every capability equally. We train an iso-parameter
4 width–depth grid of 15 masked-diffusion codec TTS models (3 seeds, 20–125 M
5 non-embedding parameters) and sweep the refinement budget T at inference only,
6 giving a 75-point (w, d, T) surface per metric from 45 trained models. Fitting
7 $\text{err} = E + Aw^{-\alpha} + Bd^{-\beta} + CT^{-\tau}$ pre-registered, test-time scaling is metric-
8 selective: $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}} = 0.110$, 95% run-level bootstrap CI [0.092, 0.131],
9 excluding zero. The effect that carries practical weight is one of *magnitude*: over
10 $T=1 \rightarrow 16$ refinement moves 0.952 absolute WER (a $6.1\times$ reduction) against
11 0.161 absolute speaker-similarity error ($1.25\times$), a $5.9\times$ asymmetry. It is not a
12 difference in convergence *rate*: on each metric’s own normalised curve the fraction
13 of total gain realised by $T=8$ is 94.6% for WER and 93.7% for identity, identity
14 marginally later. The selectivity persists across a 276 M four-budget replication,
15 $\Delta\tau = +0.1095$, CI [0.0951, 0.1241]. The asymmetry is not a limit on what
16 inference can buy, but on what *refinement* can buy: at matched cost, best-of- K
17 search moves speaker similarity by +0.0444 (CI [0.0416, 0.0470]) where refinement
18 moves it least. Depth and steps are *not* interchangeable: the substitution form
19 $B(dT^\kappa)^{-\beta}$ is worse than the separable form by $\Delta\text{AICc} = +69.3$.

20 1 Introduction

21 A diffusion language model can buy sequential computation twice. It *owns* depth: d transformer
22 blocks, paid once in parameters and then on every forward pass forever. It *rents* refinement steps:
23 T denoising passes over the same sequence, paid per utterance and adjustable at serve time. An
24 autoregressive model has only the first lever; its serial cost per token is fixed by the architecture the
25 moment training ends. If rented serial compute substitutes for owned serial compute, a deployment
26 can trade parameters for steps after the fact — but only for the capabilities that steps actually improve.

27 Zero-shot text-to-speech is an unusually clean probe of that question, because it separates two
28 capabilities that are measured by standard, independent metrics. Intelligibility is compositional: the
29 model must place the right phonetic content in the right order, and word error rate (WER) reads it
30 out. Speaker identity is representational: the model must carry timbre from a short prompt into new
31 content, and speaker similarity (SIM-o) reads that out. The two need not respond to the same knob.

32 We train one iso-parameter width–depth grid of masked-diffusion codec language models and then
33 sweep T at inference only, which yields a full (w, d, T) response surface per metric at zero additional
34 training cost. Two claims were pre-registered before any run. **H-D2**: refinement steps improve

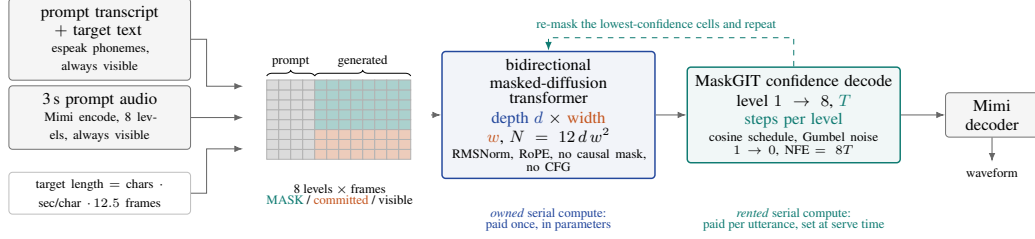


Figure 1: *The two ways a masked-diffusion TTS model buys sequential computation.* Phonemes and a 3-second prompt are always visible; the generated region of the $8 \times \text{frames}$ token grid starts fully masked. The backbone spends *depth* — *owned* serial compute, fixed at training time — and the sampler spends *T refinement steps per codebook level* — *rented* serial compute, chosen at serve time. One training grid over (w, d) therefore yields a whole (w, d, T) surface at inference. This paper asks whether the rented kind helps every capability equally.

intelligibility more than identity, i.e. test-time scaling is metric-selective. **H-D3**: for intelligibility, steps and depth exchange at a measurable rate κ — how many step-doublings buy one depth-doubling.

$$\text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m d^{-\beta_m} + C_m T^{-\tau_m} \quad \text{err}_m(w, d, T) = E_m + A_m w^{-\alpha_m} + B_m (dT^{\kappa_m})^{-\beta_m} \quad (1)$$

Both claims were registered before the first run, together with the thresholds, the frozen sampler and the analysis plan. The answer to the first is yes and quantitative: $\Delta\tau = 0.110$ with a run-level bootstrap CI of $[0.092, 0.131]$, so the step exponent is reliably larger for intelligibility than for identity. The answer to the second is a clean no: the substitution form is decisively *worse* than the separable one ($\Delta\text{AICc} = +69.3$ for WER, $+71.3$ for SIM-o) and κ runs to its bound, so no finite depth-step exchange rate describes this regime. The two results together say something sharper than either alone: extra inference compute is a targeted instrument, and it cannot be treated as a substitute for architecture.

- ❶ The first per-metric (w, d, T) scaling surface for speech generation: 75 points per metric from one training grid of 45 models, released with the harness.
- ❷ $\Delta\tau = 0.110$, CI $[0.092, 0.131]$ — steps are metric-selective — and the *negative* result on depth-step substitution stated as an AICc comparison ($+69.3$) rather than a narrative, with κ reported as unidentified.
- ❸ A serving rule measured rather than assumed: c_{layer} is flat in *width* at batch 1 on the evaluation GPU, so serial latency is $8Td_{\text{c}_{\text{layer}}}$ and the *step budget* is the only latency knob that trades against intelligibility.

2 Related work

Masked diffusion language models are now trained at scale [Nie et al., 2025, Sahoo et al., 2024, Austin et al., 2021], but their scaling is reported against a single likelihood-shaped objective, so a per-capability question cannot arise. In speech, the masked-generative recipes that made codec-token TTS practical — confidence-decoded parallel generation over residual codebook levels [Borsos et al., 2023, Chang et al., 2022] and its multi-stage successors [Wang et al., 2024] — fixed a step budget by engineering judgement and reported a single operating point; none maps error against steps as a law. Autoregressive codec TTS [Wang et al., 2023] has begun to report scaling behaviour, but in total parameters N only [Ye et al., 2025], which cannot separate width from depth and has no step axis at all; classical scaling work is likewise N -indexed [Kaplan et al., 2020, Hoffmann et al., 2022]. Test-time scaling is well studied in reasoning [Snell et al., 2024], where the payoff is measured on one correctness metric; whether extra inference compute helps different capabilities *differently* is, to our knowledge, untested in generative speech. That is the gap this abstract fills: per-metric laws in (w, d, T) for a masked-diffusion TTS model.

67 3 Method

68 **Backbone and objective.** A bidirectional (non-causal) pre-LN transformer over Mimi [Défossez
69 et al., 2024] codec tokens at 12.5 Hz with the first 8 residual codebooks (2048 entries each, plus
70 MASK and PAD). Width w , depth d , head dimension 64, GELU MLP ratio 4, RMSNorm, RoPE,
71 no weight tying, so the non-embedding parameter count is $N = 12dw^2$. One embedding table
72 per codebook is summed at the input and one linear head per codebook reads out. Training is
73 SoundStorm-style [Borsos et al., 2023] coarse-to-fine masking: per example a level $l \sim \mathcal{U}\{1..8\}$
74 and a ratio $t \sim \mathcal{U}(0, 1]$; levels below l stay visible, level l is masked i.i.d. with probability t , levels
75 above l are entirely MASK; the loss is cross-entropy on the masked cells of level l only. A phoneme
76 prefix and a 3-second prompt of codec tokens are always visible; there is no speaker encoder, so
77 the backbone itself is the only identity pathway, and no classifier-free guidance, which would add a
78 second inference-time axis and confound the step axis. The target length is fixed before generation
79 from the text length and a corpus character rate (0.0602 s/char).

80 **Frozen sampler.** MaskGIT-style confidence decoding [Chang et al., 2022], level by level, with
81 exactly T refinement steps per level, so NFE = $8T$: a cosine schedule sets how many cells remain
82 masked after each step, tokens are drawn categorically at temperature 1.0, and confidences are
83 log-probabilities plus Gumbel noise annealed $1 \rightarrow 0$ within a level; committed cells stay committed.
84 The sampler is identical for every config, seed and T — nothing is tuned per shape or per step budget.
85 Per-item generation noise is keyed so that each item sees the same noise at the first step of every level
86 for every T , making T the only difference between surface points of one run. Before any metric is
87 computed we verify that T reaches the sampler: outputs at $T=1$ and $T=16$ must differ on more than
88 20% of generated token cells.

89 **Design and statistics.** 15 configs span three iso- N budgets $\times$ five aspect ratios, each trained
90 at 3 seeds for 30,000 steps at effective batch 256 with μP [Yang et al., 2021] (base width 256,
91 base LR 0.004); width transfer and depth transfer were verified by five-point LR sweeps before
92 the grid. $T \in \{1, 2, 4, 8, 16\}$ is swept at evaluation only, giving 75 surface points per metric from
93 45 trained models. Metrics: WER (Whisper-large-v3 [Radford et al., 2023], greedy, the Whisper
94 English normaliser), SIM-o (WavLM-large [Chen et al., 2022] speaker-verification embedding
95 against the *original* prompt waveform, following Anastassiou et al. [2024]), a frozen degenerate-
96 output rule, and UTMOS [Saeki et al., 2022] as a tertiary. Error transforms are $\text{err}_{\text{WER}} = \text{WER}$
97 and $\text{err}_{\text{SIM}} = 1 - \text{SIM-o}$, computed over all items including degenerates. Both forms of Eq. (1)
98 are fitted by weighted non-linear least squares with 32 multi-starts; $\Delta\tau = \tau_{\text{WER}} - \tau_{\text{SIM}}$, κ , and
99 $\Delta\rho = \alpha_{\text{SIM}}/\beta_{\text{SIM}} - \alpha_{\text{WER}}/\beta_{\text{WER}}$ carry percentile confidence intervals from a bootstrap that
100 resamples at the *run* level — the five T -points of one run share a trained model, so resampling surface
101 points would understate their correlation — with 2000 replicates recomputing every T point.

102 4 Results

103 **Steps are metric-selective (H-D2, supported).** Fitted on all 75 surface points, the step exponent
104 is $\tau_{\text{WER}} = 0.836$ (CI [0.825, 0.851]) against $\tau_{\text{SIM}} = 0.726$ (CI [0.709, 0.744]), so $\Delta\tau = 0.110$
105 with CI [0.092, 0.131], excluding zero. The raw surface shows it without any model: over the grid,
106 $T=1 \rightarrow T=16$ takes WER from 113.8% to 18.7% (a factor of $6.1\times$ on the error scale) while SIM-o
107 moves only $0.209 \rightarrow 0.370$ ($1.25\times$ on $1 - \text{SIM-o}$), an absolute-error asymmetry of $5.9\times$. The
108 selectivity is not an artefact of scale, training compute, or the metric stack: extending the grid to a
109 fourth budget (276 M non-embedding parameters, ten further runs) gives $\Delta\tau = +0.1095$, CI [0.0951,
110 0.1241]; retraining three configurations for $3\times$ longer gives $\Delta\tau = +0.1767$, CI [0.0789, 0.2854];
111 and re-scoring every synthesis under a different ASR or refitting with unbounded log-amplitudes
112 moves it to +0.1283 and +0.1260 respectively. Extending T to 64 on a 21-run subset shows WER
113 still falling past the grid edge ($0.1516 \rightarrow 0.1294$), so T_{WER}^* on WER’s own scale is 32, confirming
114 the $T=16$ value was a lower bound. The power gate passed in all three budgets before these tests
115 were run.

116 **Depth and steps do not substitute (H-D3, not supported).** The pre-registered substitution form
117 collapses: $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$ for WER and $+71.3$ for SIM-o, where the hypothesis
118 required $\leq +4$, and the fitted κ sits on its bound for both metrics, i.e. the exchange rate is not

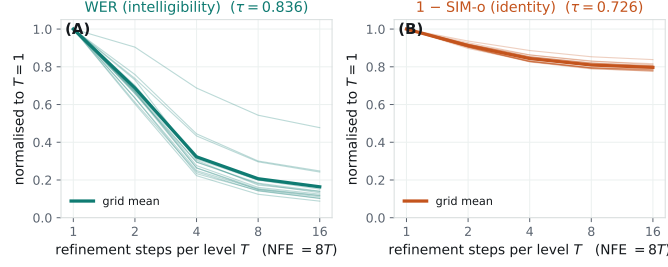


Figure 2: *Test-time scaling is metric-selective.* Error versus refinement steps for all 15 configs, each normalised to its own $T=1$ value; thick lines are grid means. (A) WER error falls 0.9518 in absolute terms over $T=1 \rightarrow 16$ and is still falling at $T=16$. (B) SIM-o error falls 0.1606 over the same range, $5.9\times$ less. Same models, same frozen sampler, same items — only T differs.

Table 1: *Fitted exponents* on the 75-point surface (M_{sep} for α, β, τ ; M_{sub} for κ) with 95% run-level bootstrap CIs for the step exponents. $\Delta\tau = 0.110$, CI [0.092, 0.131], excludes zero: the pre-registered primary result. κ is at its bound in both rows and $\Delta\text{AIC}_{\text{csub-sep}} \gg +4$, so no exchange rate is quotable. $\Delta\text{AIC}_{\text{cfull-N}} \leq -4$ means iso- N shape matters.

Metric	α (width)	β (depth)	τ (steps)	κ	$\Delta\text{AIC}_{\text{csub-sep}}$	$\Delta\text{AIC}_{\text{cfull-N}}$
WER	1.574	2.040	0.836 [0.825, 0.851]	3.000 [†]	+69.3	−15.5
SIM-o	0.919	0.553	0.726 [0.709, 0.744]	3.000 [†]	+71.3	−5.5

[†]at the fitted bound, i.e. not identified; $\Delta\tau = 0.110$, CI [0.092, 0.131].

identified. We therefore quote no κ . The mechanism is visible in the fits: T enters with a large additive term that decays as $T^{-\tau}$ towards a floor, whereas depth changes the floor itself; a model that forces the two into one factor dT^κ cannot fit both the strong depth ordering at fixed T and the strong T ordering at fixed depth.

Shape matters, but the anisotropy ratio is not identified (H-D1, not supported). At $T=16$ the two-axis model beats the N -only model for both metrics ($\Delta\text{AIC}_{\text{c}} = -15.5$ for WER, -5.5 for SIM-o, against a -4 threshold), so error is not a function of parameter count alone — iso- N shape choice matters. The direction of the anisotropy, however, we cannot claim: $\Delta\rho = 0.174$ with CI $[-0.711, 0.388]$, spanning zero, and the width amplitude A saturates the pre-registered bound $A \leq 10$ in every fit, so only $A w^{-\alpha}$ is identified and α — hence $\rho = \alpha/\beta$ — is not. The depth exponent, by contrast, *is* identified (β interior in both fits) and separates the capabilities sharply: $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} = 0.553$. What the data do show unambiguously is a strong depth ordering at fixed parameter count: moving from the shallowest shape to the best-performing depth is worth 40.2, 16.3 and 7.3 WER points in the 20, 50 and 125 M budgets, while SIM-o varies by under 0.065 across all 15 shapes. The depth optimum is *interior* only in the smallest budget ($d=18$ of 4–24); in the two larger budgets the argmin sits at the deepest shape tested ($d=30$ of 6–30 and $d=36$ of 8–36), so those are censored lower bounds and we claim no $d^*(N)$ trend.

Extrapolation and reliability. Fitting the two smaller budgets and predicting the largest gives MAPE 2.4% for SIM-o (versus 2.2% for the N -only model), inside the 15% the hypothesis asked for; WER does not extrapolate (20.2% versus 20.4%). Degenerate outputs — empty, truncated or looping transcripts under the frozen rule — fall monotonically with the step budget, from 40.4% at $T=1$ to 0.1% at $T=16$: most of what $T=1$ costs is not subtle quality but outright failures.

Serving corollary. Measured on the evaluation GPU at batch 1, c_{layer} is flat in width across the whole grid (0.493–0.514 ms per layer for $w \in [256, 1152]$): at these sizes the device is launch-bound, so serial latency is $8Td c_{\text{layer}}$ and width is free, so a latency budget constrains only the product Td . These data do not resolve how to split it: the pre-registered rule that depth should be saturated before steps are raised failed (satisfied at 5.9% of 40 log-spaced budgets by the measured method and 8.8% by the fitted one, against an 80% bar), and T is censored at 16 by the tested grid, so no directional allocation rule is claimed in either direction.

What does buy identity: search at the same cost. If refinement is an intelligibility knob, the natural question is what moves identity. At *matched* NFE we compare refinement against best-of- K sampling with candidates selected by a speaker encoder and scored by a *different*, independently validated one. At NFE=512 (refinement $T=64$ versus $K=8$ at $T=8$) search improves speaker similarity by +0.0444 (95 % CI [0.0416, 0.0470], 72 % of items), while refinement is the better spend for intelligibility (WER 0.1316 versus 0.1955). The same budget, spent two ways, buys two different capabilities. The effect is selection rather than sampling: an arbitrary candidate scores 0.4259, flat in K , so the entire gain comes from the choice. It persists at 276 M parameters (+0.0422) and at $3\times$ training compute (+0.0368), and as the baseline strengthens it claims a *larger* share of the headroom that remains to the codec ceiling (28 %, 32 %, 43 %).

Where the steps are spent matters as much as how many (exploratory). At matched total NFE = 32 on the 125 M budget, spending every refinement pass on the coarsest codebook level rather than the finest changes WER from 1.1480 to 0.2191 (paired item-level bootstrap difference -0.9289, CI [-0.9595, -0.8999]) and the degenerate rate from 39.1 % to 0.1 %; the coarse schedule is not free, costing 0.3057 against 0.3477 SIM-o and 2.17 against 2.51 UTMOS relative to a uniform split (Appendix C).

5 Limitations and outlook

The ceiling is 133 M non-embedding parameters and 30,000 steps; every model is under-trained by frontier standards, and the T curves may flatten differently at scale — $T_{\text{WER}}^* = 16$ on WER’s own scale is a lower bound, since WER had not saturated at the largest budget we measured. Part of the identity gap is not ours to close: round-tripping the ground-truth target through the codec and scoring it against the prompt gives 0.555 against 0.678 for the original audio, so 62% of the distance between real speech and our best measured system is lost in the codec itself. One language, one codec, one corpus, and no classifier-free guidance — guidance is a second inference-time axis that would confound the step axis, so excluding it is deliberate, and it is the obvious follow-up. The $\Delta\tau$ interval resamples 3 runs per config, so it captures seed-to-seed variation only, not the uncertainty of the single shared harness and corpus. The search result carries one limitation we could not close: the selector and the scorer are both WavLM/x-vector-lineage encoders (item-level $r = 0.71$), and although each independently passes the ground-truth speaker gate and the selector captures only 46 % of the gap an oracle selector would reach — where shared bias would predict nearly all of it — a genuinely independent scorer family is needed to exclude representational overlap outright. Target length is set from a corpus-median seconds-per-character rather than per-item speaking rate, so an utterance whose speaker is unusually fast or slow is synthesised at the wrong duration; this penalises both metrics and is not corrected for. $\Delta\rho$ and κ are reported as unidentified rather than estimated. Cuts taken: none. This is pre-registered outcome class S1, and selectivity is the only positive claim the data license.

Reproducibility. Protocol, hypotheses, thresholds, frozen sampler and analysis plan were registered before the first run; every gate is logged with its measured values. The released artefact holds the harness, all 45 run records, the 75-point surface, and the fitting and bootstrap code. Total cost 143 GPU-hours.

References

- Philip Anastassiou et al. Seed-TTS: A family of high-quality versatile speech generation models. *arXiv preprint arXiv:2406.02430*, 2024.
- Jacob Austin, Daniel D. Johnson, Jonathan Ho, Daniel Tarlow, and Rianne van den Berg. Structured denoising diffusion models in discrete state-spaces. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.
- Zalán Borsos, Matt Sharifi, Damien Vincent, Eugene Kharitonov, Neil Zeghidour, and Marco Tagliasacchi. SoundStorm: Efficient parallel audio generation. *arXiv preprint arXiv:2305.09636*, 2023.

197 Huiwen Chang, Han Zhang, Lu Jiang, Ce Liu, and William T. Freeman. MaskGIT: Masked generative
198 image transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*,
199 2022.

200 Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki
201 Kanda, Takuya Yoshioka, Xiong Xiao, Jian Wu, Long Zhou, Shuo Ren, Yanmin Qian, Yao Qian,
202 Michael Zeng, Xiangzhan Yu, and Furu Wei. WavLM: Large-scale self-supervised pre-training for
203 full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 2022.

204 Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou,
205 Edouard Grave, and Neil Zeghidour. Moshi: a speech-text foundation model for real-time dialogue.
206 *arXiv preprint arXiv:2410.00037*, 2024.

207 Jordan Hoffmann, Sebastian Borgeaud, Arthur Mensch, Elena Buchatskaya, Trevor Cai, Eliza
208 Rutherford, Diego de Las Casas, Lisa Anne Hendricks, Johannes Welbl, Aidan Clark, et al.
209 Training compute-optimal large language models. In *Advances in Neural Information Processing*
210 *Systems (NeurIPS)*, 2022.

211 Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child,
212 Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models.
213 *arXiv preprint arXiv:2001.08361*, 2020.

214 Shen Nie, Fengqi Zhu, Zebin You, Xiaolu Zhang, Jingyang Ou, Jun Hu, Jun Zhou, Yankai Lin, Ji-
215 Rong Wen, and Chongxuan Li. Large language diffusion models. *arXiv preprint arXiv:2502.09992*,
216 2025.

217 Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever.
218 Robust speech recognition via large-scale weak supervision. In *International Conference on*
219 *Machine Learning (ICML)*, 2023.

220 Takaaki Saeki, Detai Xin, Wataru Nakata, Tomoki Koriyama, Shinnosuke Takamichi, and Hiroshi
221 Saruwatari. UTMOS: UTokyo-SaruLab system for VoiceMOS challenge 2022. In *Interspeech*,
222 2022.

223 Subham Sekhar Sahoo, Marianne Arriola, Yair Schiff, Aaron Gokaslan, Edgar Marroquin, Justin T.
224 Chiu, Alexander Rush, and Volodymyr Kuleshov. Simple and effective masked diffusion language
225 models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

226 Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally
227 can be more effective than scaling model parameters. *arXiv preprint arXiv:2408.03314*, 2024.

228 Chengyi Wang, Sanyuan Chen, Yu Wu, Ziqiang Zhang, Long Zhou, Shujie Liu, Zhuo Chen, Yanqing
229 Liu, Huaming Wang, Jinyu Li, Lei He, Sheng Zhao, and Furu Wei. Neural codec language models
230 are zero-shot text to speech synthesizers. *arXiv preprint arXiv:2301.02111*, 2023.

231 Yuancheng Wang, Haoyue Zhan, Liwei Liu, Ruihong Zeng, Haotian Guo, Jiachen Zheng, Qiang
232 Zhang, Xueyao Zhang, Shunsi Zhang, and Zhizheng Wu. MaskGCT: Zero-shot text-to-speech
233 with masked generative codec transformer. *arXiv preprint arXiv:2409.00750*, 2024.

234 Greg Yang, Edward J. Hu, Igor Babuschkin, Szymon Sidor, Xiaodong Liu, David Farhi, Nick
235 Ryder, Jakub Pachocki, Weizhu Chen, and Jianfeng Gao. Tensor programs V: Tuning large neural
236 networks via zero-shot hyperparameter transfer. In *Advances in Neural Information Processing*
237 *Systems (NeurIPS)*, 2021.

238 Zhen Ye, Xinfu Zhu, Chi-Min Chan, Xinsheng Wang, Xu Tan, Jiahe Lei, Yi Peng, Haohe Liu, Yizhu
239 Jin, Zheqi Dai, Hongzhan Lin, Jianyi Chen, Xingjian Du, Liumeng Xue, Yunlin Chen, Zhifei Li,
240 Lei Xie, Qiuqiang Kong, Yike Guo, and Wei Xue. Llasa: Scaling train-time and inference-time
241 compute for LLaMA-based speech synthesis. *arXiv preprint arXiv:2502.04128*, 2025.

A The shape grid, as run

Every config in the active grid with its measured $T=16$ means (seeds 0, 1, 2, 400 eval_zs items each). Iso- N within a budget: the three budgets sit at 20, 50 and 125 M non-embedding parameters and the five shapes per budget span the aspect-ratio range at fixed $N = 12dw^2$.

config	budget	w	d	heads	N (M)	val loss	WER (%)	SIM-o	Degen (%)
A1	A	640	4	10	19.7	4.638	53.9	0.327	0.67
A2	A	448	8	7	19.3	4.582	28.4	0.348	0.08
A3	A	384	12	6	21.2	4.566	15.9	0.360	0.17
A4	A	320	18	5	22.1	4.574	13.6	0.359	0.00
A5	A	256	24	4	18.9	4.611	15.3	0.351	0.17
B1	B	832	6	13	49.8	4.551	27.8	0.361	0.08
B2	B	640	10	10	49.2	4.505	16.0	0.381	0.00
B3	B	512	16	8	50.3	4.497	11.9	0.388	0.08
B4	B	448	22	7	53.0	4.511	12.0	0.382	0.08
B5	B	384	30	6	53.1	4.520	11.5	0.375	0.08
C1	C	1152	8	18	127.4	4.504	17.1	0.373	0.17
C2	C	960	12	15	132.7	4.504	18.9	0.373	0.17
C3	C	768	18	12	127.4	4.488	14.3	0.392	0.00
C4	C	640	26	10	127.8	4.489	13.4	0.390	0.08
C5	C	512	36	8	113.2	4.487	9.9	0.388	0.00

B Additional figures

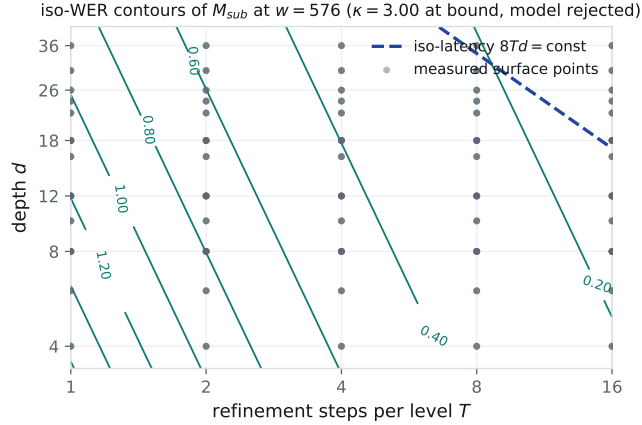


Figure 3: *Why the substitution form is rejected.* Iso-WER contours of M_{sub} in $(\log T, \log d)$ at the median width, with the iso-latency line $8Td = \text{const}$. The fitted κ sits on its bound and $\Delta\text{AICc}(M_{\text{sub}} - M_{\text{sep}}) = +69.3$, so these contours are shown as a rejected model, not as a design rule.

C Refinement-budget allocation across codebook levels

Three frozen per-level step schedules at matched total NFE = 32 on the 125 M budget (15 runs, 400 items). Paired item-level bootstrap, 2000 replicates, pairing on the eval item so item difficulty cancels. Coarse spends all passes on codebook level 0, fine on level 7, uniform splits evenly.

schedule	WER	SIM-o	UTMOS	degenerate
coarse [25, 1, ..., 1]	0.2191	0.3057	2.17	0.1%
uniform [4, ..., 4]	0.3144	0.3477	2.51	0.5%
fine [1, ..., 1, 25]	1.1480	0.2246	1.56	39.1%

Allocation is metric-selective in the same way the step *count* is: coarse wins WER but loses identity and naturalness against uniform, both with intervals excluding zero. It is a trade, not a free win.

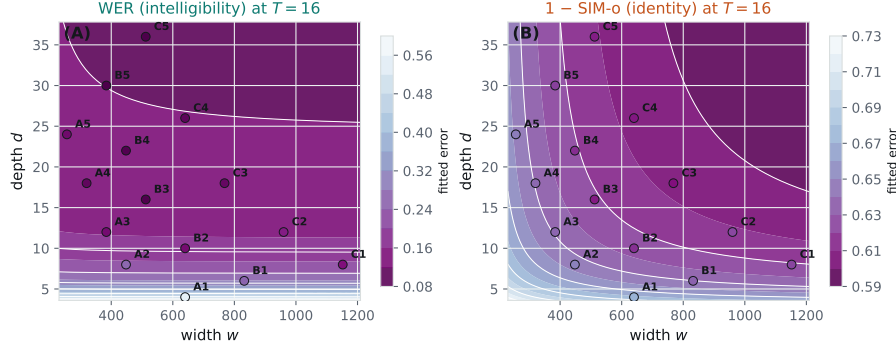


Figure 4: *Fitted anisotropy surface at $T=16$* for both metrics, with the measured config means overlaid. The width amplitude A saturates its pre-registered upper bound for both metrics, which is why $\rho = \alpha/\beta$ is reported as unidentified in §4 even though the two-axis model beats the N -only model ($\Delta\text{AICc} = -15.5$ and -5.5).

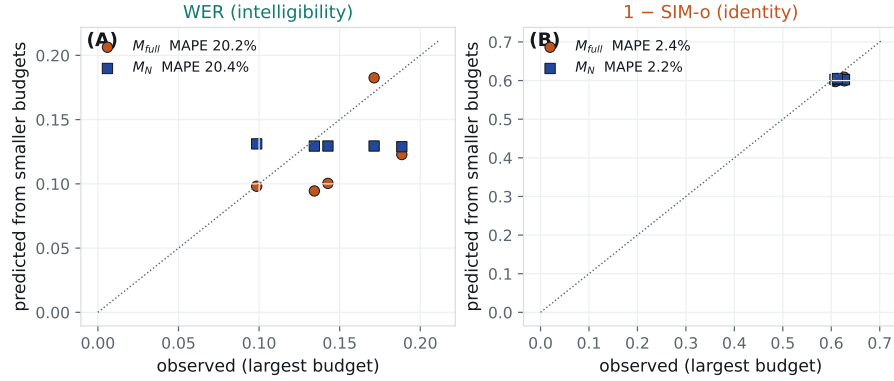


Figure 5: *H-D4 extrapolation*. The two smaller budgets are fitted and the largest budget predicted, under M_{full} and the N -only model. SIM-o extrapolates (MAPE 2.4 % vs 2.2 %); WER does not (20.2 % vs 20.4 %).

255 **A significance-based saturation contrast.** Defining saturation as the smallest T whose paired-
 256 bootstrap interval for $\text{err}(T) - \text{err}(64)$ covers zero gives $T_{\text{WER}}^* = 32$ against $T_{\text{SIM}}^* = 16$ on the
 257 extended grid. We report this only here and do not build on it: roughly half the contrast is a precision
 258 artefact, since identity’s detection floor is $\approx 1.8\times$ looser relative to its own dynamic range than
 259 intelligibility’s. The raw-scale $T^* - 95\%$ of the $T=16$ value — is not affine-invariant and must
 260 never be compared across metrics: the same 5% band is 0.67% of WER’s range but 11.96% of
 261 SIM-o’s. On the invariant fraction-of-total-gain definition, $T^* = 16$ for both metrics.

262 D Fit diagnostics and exploratory analyses

263 **Parameters at bounds.** The width amplitude A saturates the pre-registered *upper* bound $A \leq 10$ in
 264 every fit (Part A: 10.000 for both metrics; Part B: 9.999 for WER, 10.000 for SIM-o), so only the
 265 product $A w^{-\alpha}$ is identified and α is not; the bound, not the data, is binding, which is a limitation
 266 of the pre-registered parameterisation worth fixing in any follow-up. κ reaches its upper bound in
 267 every one of the 2000 bootstrap replicates of M_{sub} . The irreducible term E reaches 0 for WER under
 268 M_{sep} . The depth amplitude B is interior throughout, so β is identified.

269 **Exploratory: the depth exponent separates the capabilities.** $\beta_{\text{WER}} = 2.040$ against $\beta_{\text{SIM}} =$
 270 0.553 , a factor of 3.7 — the asymmetry H-D1 sought, in the parameter the data pin down. Part-B ρ
 271 values point the pre-registered way ($\Delta\rho = +0.890$) but inherit the A -cap problem; the pre-registered
 272 test is Part A, so this is exploratory only.

273 **Weighting.** Points are weighted $1/SE^2$ with SE taken over seeds, floored at a quarter of the pooled
274 seed SD so that a chance agreement between two seeds cannot dominate the fit; the number of floored
275 points is recorded in the released `fits.json`.

276 **Exploratory (not part of any declared claim).** An unweighted refit of both parts is released
277 alongside the weighted fit, as is a fit that additionally includes the finished runs of a stretch third seed.
278 Both reproduce the sign and significance of $\Delta\tau$.

279 **Use of Generative AI Disclosure**

280 Generative AI was used as a coding assistant for the experimental harness and for language refinement
281 of author-written text. The research design, the pre-registered hypotheses and thresholds, the analysis
282 plan, and all scientific claims are the authors' own; the authors verified the code and the numbers and
283 take full responsibility for this paper.